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SQL fundamentals

Question 1

1. `SELECT *`
`FROM employees;`

Expected output

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing
5	Eva	Maidoo	IT

2. SQL Query

`SELECT DISTINCT department`
`FROM employees;`

Expected output

department
IT
HR
Finance
Marketing

3. SQL Query
 SELECT first-name, last-name
 FROM employees
 ORDER BY salary DESC;

Expected Output

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. SQL Query
 SELECT id, first-name, last-name, salary
 FROM employees
 ORDER BY salary DESC
 LIMIT 3;

Expected Output

id	first-name	last-name	salary
5	Eva	Naidoo	72000
3	Cathy	Zulu	65000
1	Alice	Green	7000

Question 2. Filtering and where

5. SQL Query
SELECT id, first-name, last-name, department
FROM employees
WHERE department = IT;

Expected Output

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SQL Query
SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = Finance AND salary > 60000;

Expected Output

id	first-name	last-name	department
3	Cathy	Zulu	Finance

7. SQL Query
SELECT id, first-name, last-name, department
FROM employees
WHERE department = HR or department = Marketing;

Expected Output.

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SQL Query

```
SELECT id, first_name, last_name, department
FROM employees
WHERE NOT department = 'IT';
```

expected output

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SQL Query

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Expected Output

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SQL Query

```
SELECT id, first_name, last_name, department,
       salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

Expected Output

id	first_name	lastname	department
1	Alice	Green	IT
5	Eva	Naidoo	IT